

Photonic ICs Now Compete With Electronic ICs



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Electronic integrated circuits (Fig. 1) are arguably the most significant technology of the 20th century. In enabling, among other things, the computer industry, these have changed the way we work and play to an unprecedented extent.

Further, integration of nanophotonics and atomic physics has been a long-sought goal that would open new frontiers for optical physics. It is hard to argue at this stage if these kinds of integrated photonic circuits will form the innards of mass-produced devices as integrated electronic circuits have done, but it is just as hard to argue that these never will.

The original inspiration of integrated optics came from the technology of electronic integrated circuits (ICs), which has shown rapid development over several decades and has led to amazing achievements, such as complex and powerful microprocessors containing many millions of transistors, specialised signal processors and computer memory chips with huge data-storage capacities.

In combination with the all-electrical capability to control nano-scale optical circuits, one can envision very exciting opportunities for applications. Photonic inte-

grated circuits (PICs) (Fig. 2) use light rather than electrons to perform a wide variety of optical functions. Recent developments in nanostructures, metamaterials and silicon technologies have expanded the range of possible functionalities for these highly-integrated optical chips.

Integrated optics is a technology that aims at constructing so-called integrated optical devices or PICs or planar light wave circuits, containing several or many optical components that are combined to fulfill some more or less complex functions. Such components can be optical filters, modulators, amplifiers, lasers and photodetectors. These can be fabricated on the surface of some crystalline material (such as silicon, silica or LiNbO₃) and connected with waveguides.

Unfortunately, integrated optics has not been able to match the progress of microelectronics in terms of complexity of possible devices. This results from a number of technical limitations. While electronic circuits can contain extremely-small wires, optical components need to be connected via waveguides, dimensions of which usually cannot be much smaller than the wavelength, and which often cannot tolerate very sharp bends.

Optical connections between waveguides and couplers are significantly more critical than electrical connections. The ability to manipulate atoms with photons in an IC should allow physicists to explore entirely new ways in which matter and light interact.

One possibility pregnant with potential is the ability of light to manipulate and influence individual atoms. Physicists regularly use light to trap atoms and ions in the name of science. This has all kinds of important applications from quantum communication to telling the time. But devices that do all this are well beyond the reach of anybody unlucky enough not to own a well-equipped optics laboratory.

PICs could change that. These offer the possibility of using light to manipulate individual atoms in small self-contained units

Fig. 1: An electronic integrated circuit board

